Bonus: Course Enrollment Directory with `std::map`

Instructions

You must write a C++ program that simulates a simple course enrollment system.

Learning goals

- Create and use a `map<int,string>`
- Insert elements using `operator[]`
- Traverse a map using an iterator
- Access key-value pairs using `it->first` and `it->second`
- Observe that maps store keys in sorted order

Scenario

You are given a list of students enrolling in a course. Each student has a unique ID and a name.

Your task is to build and display the enrollment list.

Task

1. You must create an empty map called `enrollment` that maps student IDs to names.
2. You must add the following students using `operator[]` (in this order):
   - 3003 → "Charlie"
   - 3001 → "Alice"
   - 3004 → "Diana"
   - 3002 → "Bob"
3. You must print the enrollment list using an iterator.
4. You must number the students starting from 1 when printing.
5. You must include a comment in your code that answers this question:

- Why is "Alice" printed before "Charlie" even though she was added later?

6. You must create a second map `lateEnrollment` using an initializer list:

   - 4002 → "Evan"
   - 4001 → "Fiona"

7. You must print the second map in the same format.

Expected output

```
Enrollment list:
1. ID 3001: Alice
2. ID 3002: Bob
3. ID 3003: Charlie
4. ID 3004: Diana

Late enrollment:
1. ID 4001: Fiona
2. ID 4002: Evan
```

Starter code

```
#include <iostream>
#include <map>
#include <string>
using namespace std;

int main()
{
  // Create enrollment map

  // Add students

  // Print enrollment list with numbering

  // Create lateEnrollment map

  // Print late enrollment
```

```
  return 0;
}
```

Submission

Send me the URL to your solution in OneCompiler.com via email.